

MEMBERS (1-34 of 102)

Mark	Section	L m	n	kg	between
T2	IPE 240	4.07	1	125	R78-R82
P14	IPE 300	4.07	1	172	R78-K21
R1.1	IPE 240	5.89	1	181	K25-K19
R1.2	IPE 240	4.61	1	142	K19-K15
R1.3	IPE 240	3.00	1	92	K15-K8
R1.4	IPE 240	5.80	1	178	K8-K6
R2.1	IPE 240	5.89	1	181	P_K25K26-P_K19K20
R2.2	IPE 240	7.56	1	232	P_K19K20-P_K8K9
R2.3	IPE 240	5.90	1	181	P_K8K9-P_K6K5
R3.1	IPE 240	5.89	1	181	K26-P_K19K20
R3.2	IPE 240	7.51	1	231	P_K19K20-K9
R3.3	IPE 240	6.00	1	184	K9-K5
R4.1	IPE 240	5.79	1	178	K24-P_K19K20
R4.2	IPE 240	7.51	1	231	P_K19K20-P_K9K10
R4.3	IPE 240	5.93	1	182	P_K9K10-P_K5K7
R5.1	IPE 240	5.89	1	181	K27-K20
R5.2	IPE 240	2.70	1	83	K20-K16
R5.3	IPE 240	4.81	1	148	K16-K10
R5.4	IPE 240	5.90	1	181	K10-K7
R6.1	IPE 240	4.39	1	135	K21-K17
R6.2	IPE 240	4.81	1	148	K17-K11
R6.3	IPE 240	6.40	1	196	K11-K3
R7.1	IPE 240	9.20	1	282	P_K21K22-P_K11K12
R7.2	IPE 240	6.40	1	196	P_K11K12-P_K3K1
R8.1	IPE 240	9.20	1	282	P_K21K22-K12
R8.2	IPE 240	6.50	1	200	K12-K1
R9.1	IPE 240	9.20	1	282	P_K22K23-P_K12K13
R9.2	IPE 240	6.50	1	200	P_K12K13-P_K1K2
R10.1	IPE 240	9.20	1	282	P_K22K23-K13
R10.2	IPE 240	6.50	1	200	K13-K2
R11.1	IPE 240	4.39	1	135	K23-K18
R11.2	IPE 240	4.81	1	148	K18-K14
R11.3	IPE 240	6.40	1	196	K14-K4
P1	IPE 300	4.10	1	173	K6-K5

MEMBERS (35-68 of 102)

Mark	Section	L m	n	kg	between
P2	IPE 300	5.69	1	240	K5-K7
P3	IPE 300	5.34	1	225	K3-K1
P4	IPE 300	5.29	1	223	K1-K2
P5	IPE 300	3.07	1	130	K2-K4
P6	IPE 300	4.10	1	173	K8-K9
P7	IPE 300	5.69	1	240	K9-K10
P8	IPE 300	4.07	1	172	K10-K11
P9	IPE 300	5.34	1	225	K11-K12
P10	IPE 300	5.29	1	223	K12-K13
P11	IPE 300	3.07	1	130	K13-K14
P12	IPE 300	4.07	1	172	K16-K17
P13	IPE 330	9.79	1	481	K19-K20
P15	IPE 300	7.03	1	297	K21-K22
P16	IPE 300	6.67	1	281	K22-K23
P17	IPE 300	4.00	1	169	K25-K26
P18	IPE 300	2.90	1	122	K26-K24
P19	IPE 300	2.89	1	122	K24-K27
C1	HEA 140	3.00	1	74	K1
C2	HEA 140	3.00	1	74	K2
C3	HEA 140	3.01	1	74	K3
C4	HEA 140	3.01	1	74	K4
C5	HEA 140	3.03	1	75	K5
C6	HEA 140	3.04	1	75	K6
C7	HEA 140	3.04	1	75	K7
C8	HEA 140	3.38	1	83	K8
C9	HEA 140	3.39	1	84	K9
C10	HEA 140	3.39	1	84	K10
C11	HEA 140	3.39	1	84	K11
C12	HEA 140	3.39	1	84	K12
C13	HEA 140	3.39	1	84	K13
C14	HEA 140	3.39	1	84	K14
C15	HEA 140	3.56	1	88	K15
C16	HEA 140	3.68	1	91	K16
C17	HEA 140	3.68	1	91	K17

MEMBERS (69-102 of 102)

Mark	Section	L m	n	kg	between
C18	HEA 140	3.68	1	91	K18
C19	HEA 140	3.81	1	94	K19
C20	HEA 140	3.81	1	94	K20
C21	HEA 140	3.94	1	97	K21
C22	HEA 140	3.94	1	97	K22
C23	HEA 140	3.94	1	97	K23
C24	HEA 140	4.19	1	103	K24
C25	HEA 140	4.20	1	104	K25
C26	HEA 140	4.20	1	104	K26
C27	HEA 140	4.20	1	104	K27
WP1	HEA 140	4.24	1	105	notch corner
ST1	IPE 240	5.65	1	173	R9-R11
ST2	IPE 240	4.10	1	126	R1-R3
B1	L 70x7	6.19	2	91	K1-K2
B2	L 70x7	6.56	2	97	K5-K7
B3	L 70x7	7.81	2	115	K22-K23
B4	L 70x7	5.96	2	88	K25-K26
B5	L 70x7	6.05	2	89	K15-K19
B6	L 70x7	6.10	2	90	K14-K18
B7	L 70x7	7.26	2	107	K20-K27
B9	L 70x7	5.96	2	88	K18-K23
RT-N-W.1	M24 rod 8.8	7.18	2	51	panel x 68.0-72.1
RT-N-W.2	M24 rod 8.8	8.22	2	58	panel x 72.1-77.8
RT-N-E.1	M24 rod 8.8	8.37	2	59	panel x 81.8-87.2
RT-N-E.2	M24 rod 8.8	8.38	2	59	panel x 87.2-92.5
RT-N-E.3	M24 rod 8.8	7.10	2	50	panel x 92.5-95.5
RT-JOG.1	M24 rod 8.8	6.30	2	45	panel x 77.8-81.8
RT-W.1	M24 rod 8.8	7.18	2	51	panel x 68.0-72.1
RT-W.2	M24 rod 8.8	6.17	2	44	panel x 68.0-72.1
RT-W.3	M24 rod 8.8	5.05	2	36	panel x 68.0-72.1
RT-W.4	M24 rod 8.8	7.18	2	51	panel x 68.0-72.1
RT-E.1	M24 rod 8.8	7.16	2	51	panel x 89.9-95.5
RT-E.2	M24 rod 8.8	7.42	2	53	panel x 89.9-95.5
RT-E.3	M24 rod 8.8	8.57	2	61	panel x 89.9-95.5

COLD-FORMED AND TOTALS BY SECTION

Item	Section	Qty	Total m	kg
Hot-rolled / bracing members above	IPE 300	18	82.7	3489
Hot-rolled / bracing members above	IPE 330	1	9.8	481
Hot-rolled / bracing members above	IPE 240	34	204.3	6272
Hot-rolled / bracing members above	HEA 140	28	99.9	2468
Hot-rolled / bracing members above	L 70x7	16	103.8	766
Hot-rolled / bracing members above	M24 rod 8.8	26	188.6	669
Purlins Z200x2.0 @ 1.5 m (110 pcs) + anti-sag	Z200x2.0	110	290.2	1798
Girts Z200x2.0 (58 pcs) incl. sleeves	Z200x2.0	58	281.2	1792
Eave rails C200 (G-W 10.0 + G-E 13.9) + well header 2 x 3.9	C200x60x2.5	4	31.7	219
Sill rail C100x50x3 (D11)	C100x50x3	-	101	404
Upstand framing C100x50x3 (both wells)	C100x50x3	-	90	360

TOTALS

Hot-rolled members 12.7 t + plates 2.3 t = 15.1 t
Wall bracing + roof rods 1.44 t; cold-formed Z/C 4.6 t
STEEL WEIGHT 21.1 t (43 kg/m2 of 486 m2 footprint, 48 kg/m2 of 439 m2 roofed)
Design report Rev 6a s.10: BOM basis 22.1 t (P13 IPE 330 + 70 kg).
Galvanising: hot-dip all members and plates; Z/C S350GD Z275. Bases per bases_C.md Rev 8 (type E, P at K21).

PLATES - estimate

Plates and fittings (S275 unless noted)	Qty	kg
Fin plates FP1 100x150x10 (D1)	53	62.4
Fin plates FP2 160x150x10 (D1, chord splices)	16	30.1
Cap plates 200x280x20 (D2)	27	237.4
Tie plates 140x400x10 (D3, rows F and B)	8	35.2
Base plates 300x400x20 (type E x13 + K21)	14	263.8
Key-pair plates 800x400x20 (K1, K2, K5, K10, K22)	5	251.2
Key-pair plates 400x800x20 (K15, K18, K19, K20)	4	201.0
Key-pair plates K7 400x950, K23 1000x400, K25 / K27 400x850 x20	4	226.1
Key A SHS 90x90x8 x 205 S355 (14 + 13 pairs)	40	166.1
Key B dia 60 x 205 S355 (K3, K4 x2, K6 x2, K8, K14) + WP1	8	36.4
K21 saddle plates 400x150x15	2	14.1
Bracing gussets 10 mm ~350x420 (D4)	32	221.6
Rod gussets 8 mm ~250x250 (D5, 13 panels)	52	122.5
Purlin cleats 120x160x8 (D8)	220	265.3
Girt cleats L100x100x8 x150 (D9)	116	204.9
WP1 base plate 250x250x15 + head cleat	1	9.2

BOLTS AND ANCHORS - estimate

Bolts, bars and anchors (add 5 % spare)	Qty
M20 8.8 x 60 (fin plates FP1 / FP2)	168
M20 8.8 x 70 (cap + tie plates)	108
M20 8.8 x 60 (bracing gussets, 2 per angle end)	64
M16 8.8 (rod gussets 2 per rafter leg; crossing clips)	112
M24 8.8 rods with turnbuckle, nuts + lock nuts	26
M12 8.8 (purlin / girt cleats, fly braces, eave rail)	872
dia 16 B500 post-installed bars M16 threaded, EAD 330087 injection (27 bases x 4)	108
M12 anchors (WP1 location)	2
M8 resin anchors sill rail @ 600 (D11)	170
Key A pockets 140 x 200 cored; Key B pockets 110 x 200	48
Pilot holes 10 x 600 + production holes 20 x 600 (bars)	108

PROJECT	SHEET TITLE	REVISIONS	CURRENT REVISION	DRAWING No.
Steel roof - Administration Building, Regatta Tourist Village, Tripoli	BILL OF MATERIALS	32026-09-29Rev 6a (IPE 300/240, HEA 140), bases Rev 8, bracing Rev 5a	Rev 6a superstructure Rev 8 bases / bracing Rev 5a	RTV-ST-S06
ALTERNATIVE C - POST-AND-BEAM BRACED STEEL ROOF, SANDWICH PANELS	Lengths in m from members_C.csv Rev 6a; weights nominal (kg/m x length), plates by volume x 7850	22026-09-28Design Rev 3 / bases Rev 4b (critique C1-C15)	DATE 2026-09-29	S06
Client: Regatta Tourist Village, Tripoli Issued by: Eng. MALEK ABOZRAIG (malek.abozraig@gmail.com) Units: m	Status: ISSUED FOR CONSTRUCTION subject to the site verification of S00 section 9	12026-09-27First issue: design Rev 2, bases Rev 3	Drawn: Designed: Checked: Approved:	